

Use PostgreSQL Upsert in Python



Upsert



by

Bryan Cafferky

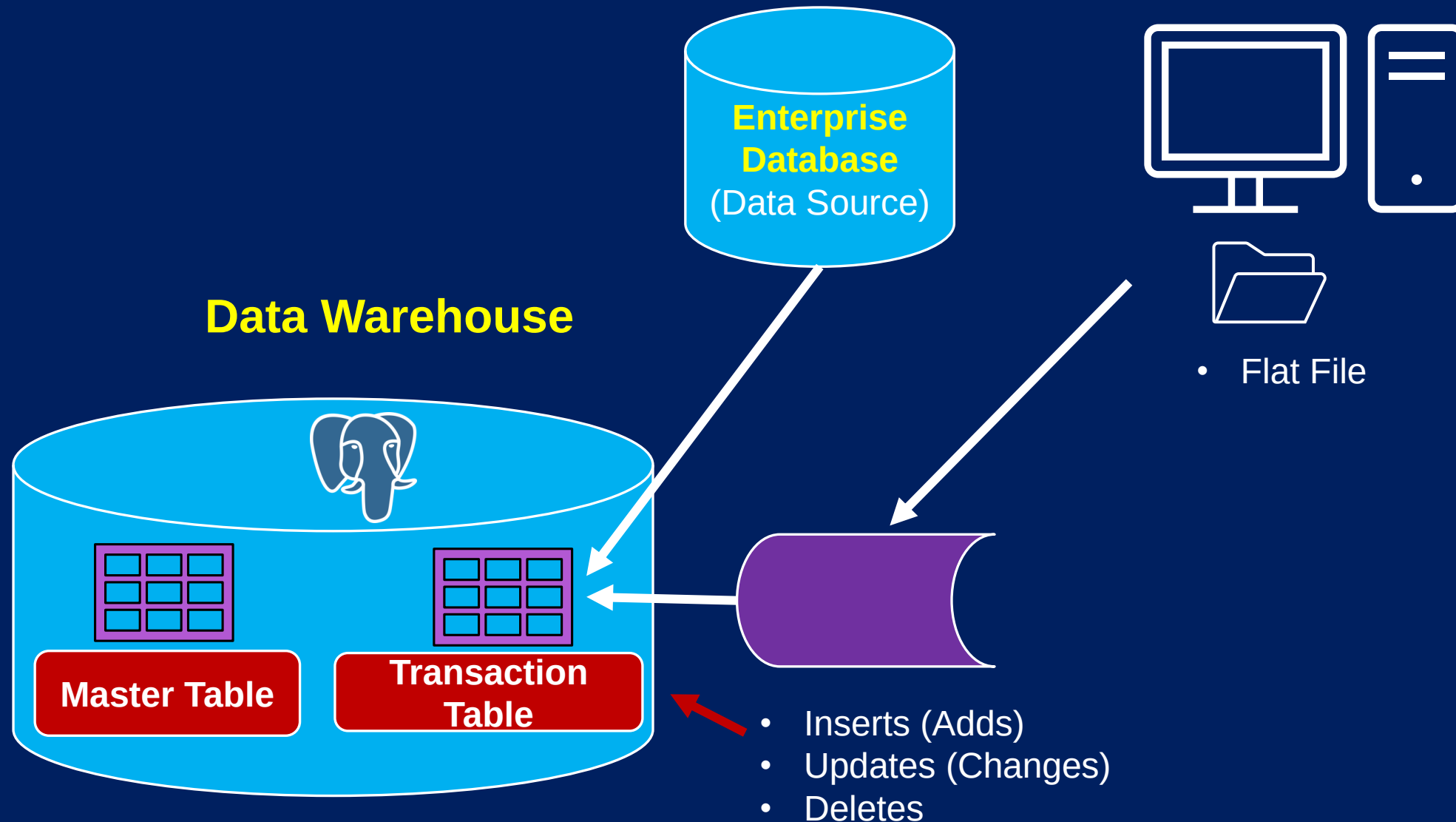
Data and AI Enabler

Where Are We Going?



- **The Problem: Keeping Data Current**
- **What is an Upsert & Why Do You care?**
- **The Tools of the Trade**
- **Riveting Demo**

Problem: Database Table Maintenance



Traditional Solution



- Use Insert Statement to Perform Adds.
- Update Statement to Apply Changes.
- Delete Statement to Remove Rows.



Upsert

What is an Upsert?



- **SQL Statement that Supports Both Inserts and Updates.**
- **Leverages a Table's Unique column to Determine Whether to Apply an Insert or an Update.**
- **Support for Conditional Logic.**

Why Do I Care?



- **Makes Your Life Easier!!!**
- **Good to Be Aware Of**
- **Another Tool in your Toolbox**

The Tools of the Trade



pgAdmin 4

127.0.0.1:61136/browser/

pgAdmin

Browser

Servers (1)

- PostgreSQL 13
 - Databases (3)
 - aw
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Schemas (2)
 - public
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries

Properties SQL Statistics Dependencies Dependents

☐	Name	Owner	Partitioned table?	Comment
☐	customer	postgres	☐ False	
☐	customerx	postgres	☐ False	
☐	dimdate	postgres	☐ False	
☐	product	postgres	☐ False	
☐	sales	postgres	☐ False	
☐	youtube.customer	postgres	☐ False	
☐	youtube_customer	postgres	☐ False	
☐	youtube_custtrans	postgres	☐ False	

Watch My
Other
PostgreSQL
Videos

<https://www.pgadmin.org/docs/pgadmin4/latest/index.html>

Demo



Demo

Wrapping Up



- **The Problem: Keeping Data Current**
- **What is an Upsert & Why Do You care?**
- **The Tools of the Trade**
- **Riveting Demo**

Thank You!